

Euan Hughes

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WORK EXPERIENCE

Researcher and Engineer Intern | Autodesk

JAN 2026 – MAY 2026

- Added raytraced, relightable 3D Gaussian splats to the Arnold rendering core, on both CPU (SIMD) and GPU (OptiX); **shipped in Arnold 7.5.2 and 3ds Max 2027**.
- Derived novel splat-intersection methods, numerically stable and **2.6× faster** than naive intersection, that **outperformed competing production renderers**.

Research Assistant | University of Waterloo

SEP 2024 – APR 2025

Supervisor: *Prof. Toshiya Hachisuka*

- Developed the core statistical formulation behind a neural spatiotemporal denoiser for Monte Carlo rendering animations (second author, **Eurographics 2026**).
- Implemented a common-random-number animation integrator in Mitsuba to generate network inputs; **cut inference time by >99.9%** via vectorization and parallelization.

Infantry Soldier | Canadian Army Primary Reserve

MAY 2019 – FEB 2023

EDUCATION

École de Technologie Supérieure

MAY 2025 – APR 2027 (*Expected*)

Master of Applied Science in Computer Science, GPA: 4.3/4.3 (perfect)

Montréal, QC

Supervisors: *Prof. Adrien Gruson* and *Prof. Eric Paquette*

- Coursework: **Photorealistic Rendering Algorithms (A+)**, **Polygon Mesh Processing (A+)**

University of Waterloo

SEP 2020 – DEC 2024

Bachelor of Computer Science

Waterloo, ON

- Coursework: **Computer Graphics (99)**

PUBLICATIONS

Deep Residual Combiner: A Learned Fusion of Spatial, Temporal, and Multiscale Correlated Pixel Estimates

2026

Eurographics 2026 · doi:10.1111/cgf.70358

Weijie Zhou, **Euan Hughes**, Toshiya Hachisuka

PROJECTS & OPEN SOURCE

More projects at wantonsushi.github.io

mr-sanders – production-style Monte Carlo renderer | C++

- From-scratch path-tracing core with MIS, light-tree sampling for many-light scenes, QMC sampling, and a guided photon-mapping solver that resolves caustics on difficult (SDS) paths.
- Production-style shading and assets: OSL/OpenPBR/MaterialX materials, hair, and heterogeneous OpenVDB volumes, with Embree traversal, multi-level instancing, motion blur, and an interactive progressive preview.
- ACEScg color management, AOVs, adaptive sampling, OIDN denoising, and developer tooling (diagnostics/test/profiling setup).

Metropolis Light Transport Visualizer | JavaScript, WebGL2

- In-browser Metropolis Light Transport renderer that visualizes Markov-chain path proposals, accepts/rejects, and image convergence in real-time.

Research Rendering Engine | C++

- Won the ÉTS rendering competition.** Rendering research testbed implementing advanced algorithms including Metropolis, gradient-domain, and bidirectional methods. Actively maintained for fast prototyping of rendering algorithms.

OpenQMC – Academy Software Foundation quasi-Monte Carlo sampling library | C++

Contributor

TECHNICAL SKILLS

Languages: C++ (11–20), C, CUDA, Python, JavaScript
Graphics APIs: OptiX, Vulkan, WebGL2, Embree
Tools: Git, Profiling, PyTorch, Blender

SCHOLARSHIPS AND AWARDS

(\$5,000): ÉTS Internal Scholarship	MAR 2026
1 st place: ÉTS Rendering Competition	DEC 2025
(\$1,500): President’s Research Award	SEP 2024
(\$3,000): President’s Scholarship of Distinction	SEP 2020